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Full Length Article

Multi-source domain open-set deep transfer adversarial network for operating performance assessment ☆

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ABSTRACT

The process operating performance assessment (POPA) of electro-fused magnesium furnace (EFMF) is very important to ensure product quality and pursue the maximum comprehensive economic benefit. However, the data at the beginning of the new production processes do not have performance grade labels and often includes new performance grades. Traditional multi-source domain open-set domain adaptation (OSDA) method categorizes all unknown classes into one class without further subdivision. To address this issue, a method based on multi-source domain open-set deep transfer adversarial network (MDODTAN) is studied to solve the POPA problem of the EFMF, which focuses on subdividing multiple unknown classes into different unknown performance grades. This network designs a task classifier for each source domain, and the assessment accuracy of known performance grades is further enhanced. Then, the domain gap between the known performance grades in each source-target domain is reduced through multi-source domain adversarial training. By constructing a similarity matrix between the known and unknown performance grades, pseudo-labels are assigned to the target domain data, and the assessment accuracy of performance grade of the new smelting process is improved through iterative training. The experimental results indicate that our method achieves higher performance assessment accuracy in open-set scenarios compared to existing methods, while also accurately classifying and subdividing multiple unknown performance grades.

1. Introduction

As an important refractory material, fused magnesia has the characteristics of high melting point, oxidation resistance and corrosion resistance, and it has broad application prospects in aerospace, electronic equipment, chemical industry and other fields. A special three-phase alternating current electro-fused magnesia furnace (EFMF) (Chai et al., 2021; Li et al., 2018; Wu et al., 2015) is a key equipment for producing high purity fused magnesia. In the actual production process, new working conditions arising from fluctuations in raw material properties or new production lines resulting from equipment upgrades and transformations can be referred to as “new production processes.” The data at the beginning of the new production processes do not have performance grade labels and often includes new performance grades. Therefore, in order to maximum comprehensive economic benefit of the product and ensure the best operating performance, the process operating performance assessment (POPA) is of great significance especially for the smelting process of EFMF with new performance grades.

POPA usually divides the operating performance into several grades, such as optimal, suboptimal, general, poor and so on based on expert experience and comprehensive economic indicators. In recent years, scholars initially used traditional multivariate statistical analysis, machine learning and other methods to build POPA models (Chu et al., 2024; Ye et al., 2009; Zhang et al., 2024; Zou et al., 2020). A POPA method based on slow feature analysis and Gaussian process regression (SFA-GPR) to obtain more complete feature information was proposed in Wang et al.

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(2022). Furthermore, Zou *et al.* extracted process data features using a combination of canonical variable analysis and slow feature analysis (Zou & Zhao, 2020). Fan *et al.* proposed a decentralized POPA scheme using multi-block total projection to latent structures (MB-T-PLS) and Bayesian inference for geological drilling (Fan et al., 2022). The above POPA method only extracts features once, while many POPA methods based on deep learning have been proposed to dig deep features of performance grades. Since most industrial processes are highly nonlinear and dynamic, a POPA network was proposed in Chu et al. (2023). For hot strip mills with multiple subsystems, Zhang et al. (2023) proposed a distributed Siamese gated recurrent unit (DSGRU) network. Furthermore, Bu *et al.* solved the POPA problem of inconsistent distribution of training set and test set (Bu et al., 2024). The POPA methods mentioned above have achieved satisfactory results in specific industrial applications. However, they all rely on labeled data and struggle to address the issues of unlabeled data and new performance grades in new production processes.

Within the research context of POPA, with historical production process data exhibiting similar characteristics designated as the source domain (SD) and data from new production processes termed the target domain (TD), a central challenge lies in effectively transferring performance grade features learned from the labeled SD to the unlabeled TD for performance grade identification. This transfer strategy not only addresses the rapid changes and dynamic optimization demands inherent in industrial environments but also enhances the generalization capability and adaptability of POPA models. Inspired by transfer learning, domain adaptation (DA) (Cai et al., 2024, 2026; Liu et al., 2025; Ning et al., 2025; Tian et al., 2024) has become a research hotspot. It aims to reduce the distribution gap between SD and TD, enabling models trained on the SD to perform well on the TD. Yaroslav *et al.* proposed a domain adversarial neural network (DANN), which learns domain invariant features of SD and TD data (Ganin et al., 2016). Then, Eric *et al.* combined adversarial learning methods and proposed a adversarial discriminative domain adaptation (ADDA) method (Tzeng et al., 2017). For the situation where labeled samples come from multiple domains, a multi-SDs adaptive transfer learning method was proposed in Zhu et al. (2021). Cai *et al.* proposed a causal-guided adaptive multimodal diffusion networks (C-AMDN) to address the problem of distribution alignment and semantic information preservation between source and target domains in multi-source DA (Cai et al., 2025). Nevertheless, these methods can only handle cases where the SD and TD contain the same performance grade, which we refer to as “known performance grade,” and cannot handle single or multiple new performance grades in the TD that do not appear in the SD, which we refer to as “unknown performance grade.” To address this issue, open-set domain adaptation (OSDA) (Alvarenga e Silva et al., 2025; Tian et al., 2023b, 2025b) is receiving increasing attention. Saito *et al.* proposed an OSDA by backpropagation (OSDAB) method that filters unknown samples (Saito et al., 2018). Liu *et al.* proposed an approach-separate to adapt (STA), gradually distinguishing unknown and known samples (Liu et al., 2019). For open-set fault diagnosis without extra fault labels, a multi-adversarial learning DA network was proposed in Zhu et al. (2023). Single-SD often fails to comprehensively reflect the performance of model, leading to biased or unrepresentative assessment results. Therefore, introducing multi-SDs can effectively enhance generalization ability and assessment reliability. A multi-source weighted deep transfer network was proposed for open-set fault diagnosis (Chen et al., 2023). Tian *et al.* proposed an open-set multi-source DA method for practical engineering applications (Tian et al., 2023a). For multi-source open-set fault diagnosis with category shift, a multi-adversarial deep transition network (MADTN) was proposed in Zuqiang et al. (2023). However, significant differences exist among different SDs, along with issues such as inconsistent label space structures and potential noise interference. This makes the training process and assessment strategies for multi-SDs challenging. The above methods, whether single-SD or multi-SDs, ultimately classify all unknown classes into a single category. To overcome this limitation, a minority of innovative studies have begun to explore more advanced solutions, attempting to perform fine-grained segmentation of unknown classes. Tan *et al.* proposed a universal DA based on reweighted optimal transport (UDA-ROT) for cross-domain fault diagnosis tasks of large-scale systems (Zeng et al., 2025). Tong *et al.* achieve class classification by mining inter-class relationships in OSDA, reserving space for unknown classes (Tong et al., 2025). Furthermore, a source-free domain adaptation for open-set (SFDA-OS) approach was proposed to address cross-domain fault diagnosis under privacy-preserving scenarios (Tian et al., 2025a). Despite advancements in handling unknown classes, a deeper analysis reveals that neither unifying unknown classes into a single category nor segmenting them has successfully addressed the open-set POPA problem.

Based on the above-mentioned observations, in order to accurately classify multiple unknown performance grades, we propose a multi-source domain open-set deep transfer adversarial network (MDODTAN) to solve the open-set POPA problem of EFMF. To avoid misclassifying known performance grades that only exist in individual SDs, we design an independent task classifier for each SD. Information entropy is used to separate the known and unknown performance grade datasets in the TD, with lower entropy indicating more stable and reliable assessment results. Through adversarial training in each source-target domain, the classification accuracy for known performance grades is improved. Then, pseudo-label prediction is performed on the TD data using similarity matrix and comprehensive voting method, and high-precision assessment of the performance grade of the new smelting process is achieved through iterative training. The contributions of this research are summarized as follows:

1) The MDODTAN method breaks away from the traditional OSDA approach of simply grouping all unknown classes into one category without further subdivision. It is capable of further subdividing multiple unknown performance grades, making it more relevant to real-world applications.

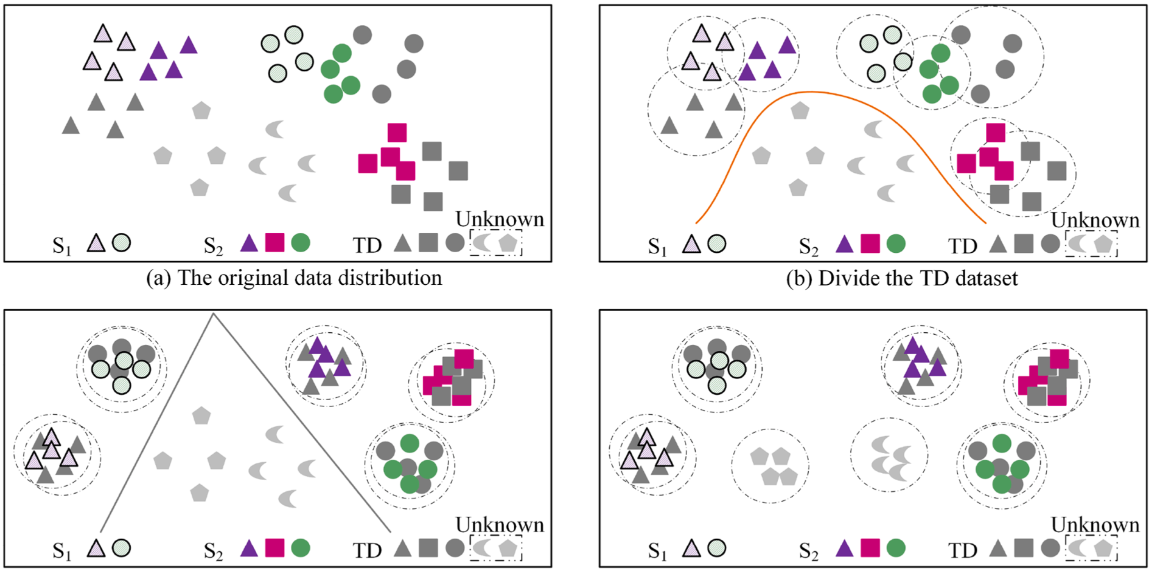
2) Each SD has its own independent task classifier, based on the fact that each SD contains a different number of performance grades. This effectively suppresses the negative transfer phenomenon that can be caused by a shared classifier across multiple SDs. Experimental results demonstrate that our method significantly improves the classification accuracy of known performance grades in the TD, enabling faster knowledge transfer and model convergence.

3) To avoid misclassifying known performance grades that only exist in individual SDs as unknown performance grades, the MDODTAN method abandons the direct prediction of pseudo-labels. Instead, it employs a comprehensive voting method to integrate the assessment results of each classifier on known performance grades. The similarity matrix is then utilized to assign accurate pseudo-labels to the target samples, further enhancing the accuracy and reliability of the assessment.

The rest of this article is organized as below. Section II introduces the proposed MDODTAN method. In Section III, the validity of the proposed method is verified by the simulation data of EFMF. Finally, conclusions are made in Section IV.

2. Proposed method

In this section, we present an overview of the MDODTAN method and then describe the training process in detail. Fig. 1 shows the overall data trend of the proposed MDODTAN method during offline training, gray shapes are data of TD and shapes in color are data of SD. Different kinds of shapes represent different performance grades. It is divided into four parts. Part I is the original data distribution, where TD have unknown performance grades. Part II effectively separates known and unknown performance grades in TD by using information entropy. Part III is the situation after adversarial training for each source-target domain. Part IV accurately classifies multiple unknown performance grades in the TD.



(a) The original data distribution (b) Divide the TD dataset (c) Adversarial training (d) Predict pseudo-labels for TD samples

Fig. 1. The overall data trend of the proposed MDODTAN method during offline training.

Table 1
Current variables of the EFMF.

No.	Variable Descriptions	Units
1	Mean of distance from A-phase electrode to liquid surface	mm
2	Mean of distance from B-phase electrode to liquid surface	mm
3	Mean of distance from C-phase electrode to liquid surface	mm
4	Variance of distance from A-phase electrode to liquid surface	—
5	Variance of distance from B-phase electrode to liquid surface	—
6	Variance of distance from C-phase electrode to liquid surface	—
7	Mean of A-phase electrode current	A
8	Mean of B-phase electrode current	A
9	Mean of C-phase electrode current	A
10	Variance of A-phase electrode current	—
11	Variance of B-phase electrode current	—
12	Variance of C-phase electrode current	—

2.1. Problem description

According to the process mechanism of the EFMF, current information is crucial for reflecting smelting process performance. Analyzing its characteristics under different operating conditions is essential for accurate performance classification. This study includes 12 current variables, as shown in Table 1.

Some necessary symbols and concepts will be used in this paper. The multi-SDs data are from the dataset of M different EFMFs in the historical melting processes. Denote the SD $S_m, m = 1, 2, \dots, M$ dataset as $D_{S_m} = \{\mathbf{x}^i_{S_m}, y^i_{S_m}\}_{i=1}^{n_{S_m}}$ with n_{S_m} labeled samples. The TD data are taken from the new smelting processes of EFMF. Denote the TD dataset as $D_T = \{\mathbf{x}^j_T\}_{j=1}^{n_T}$ with n_T unlabeled samples. For label setting, C_{S_m} denotes the source label set of S_m and C_T denotes the target label set. The number and types of known performance grades contained in different SDs vary. For S_m, $C_{S_m} = \{y^i_{S_m} = k | k = 1, 2, \dots, K_{S_m}\}_{i=1}^{n_{S_m}}$ has K_{S_m} performance grades, where $y^i_{S_m} = k$ indicates the i th sample $\mathbf{x}^i_{S_m}$ of the k th performance grade in S_m. The source label set $C_S = \bigcup_{m=1}^{M} C_{S_m} = \{1, 2, \dots, K\}$ is a union label set of all SDs with a total of K known performance grades. In this article, C_T contains all the known performance grades that have already appeared in C_S. In addition, it also includes one or more unknown performance grade. In previous OSDA studies, researchers generally classified all unknown classes into a simple class, that is, the $(K + 1)$ th class. In order to further accurately distinguish multiple unknown performance grades, the number of performance grades in the TD is set to $K + K_U$, where K_U represents the number of unknown performance grades. The online test samples are taken from the TD dataset and represented as $X_{on} = \{\mathbf{x}^j_{on}\}_{j=1}^{n_{T_{on}}}$. According to the hypothesis of typical OSDA method (Saito et al., 2018; Zuqiang et al., 2023), the probability distributions of each performance grade data in multi-SDs and TD are different, i.e., when $m \neq n$, $P_{S_m}(X_{S_m}) \neq P_{S_n}(X_{S_n})$ and $P_{S_m}(X_{S_m}) \neq P_T(X_T)$. Let X_{S_m} and X_T represent the feature spaces of samples in S_m and TD, respectively. The purpose of this study is to propose the MDODTAN method, which not only classifies multiple unknown performance grades in the TD, but also subdivides them into different unknown performance grades.

2.2. Structure of MDODTAN

In order to solve the problem of open-set POPA for new production processes, we propose an operating performance assessment method based on MDODTAN in this section. The structure of the MDODTAN is shown in Fig. 2. The MDODTAN model consists seven parts: shared feature extractor $F_{sfe}(\mathbf{x})$, multi-SD feature consistent module $F_{mdc}(\mathbf{x})$, task classifier $F^m_c(\mathbf{h}), m = 1, \dots, M$, TD data division module $F_{tdd}(\mathbf{a})$, domain discriminator $F_{dd}(\mathbf{h})$, comprehensive assessment module $F_{ca}(\mathbf{a})$, and similarity measure module $F_{sm}(\mathbf{x})$. ψ_{sfe}, ψ_{mdc}, $\psi^m_c (m = 1, \dots, M)$, ψ_{tdd}, ψ_{dd}, ψ_{ca}, ψ_{sm} are their corresponding parameters. $\mathbf{x}$ represents the input sample, $\mathbf{h}$ is the deep feature extracted by the feature extractor, and $\mathbf{a}$ is the operating performance assessment result.

(1) Shared feature extractor

Shared feature extractor is used to extract deep features of current information in multi-SDs and TD. According to the characteristics of current data, a two-layer full connect neural network (FCNN) with hidden nodes of 10 and 8 is used.

(2) Multi-SD feature consistent module

In order to ensure that cross-domain data of the same performance grades aggregate in the feature space and expand the difference between different performance grades, the idea of minimum intra-class distance and maximum inter-class distance is adopted.

The intra-class distance is defined as follows:

$$L_{intra} = F_{mdc}(\mathbf{x}) = \sum_{k=1}^{K} \sum_{i=1}^{N_{D_k}} \left\| F_{sfe}(\mathbf{x}^i_k) - \bar{F}_{sfe,k} \right\|^2, \quad (1)$$

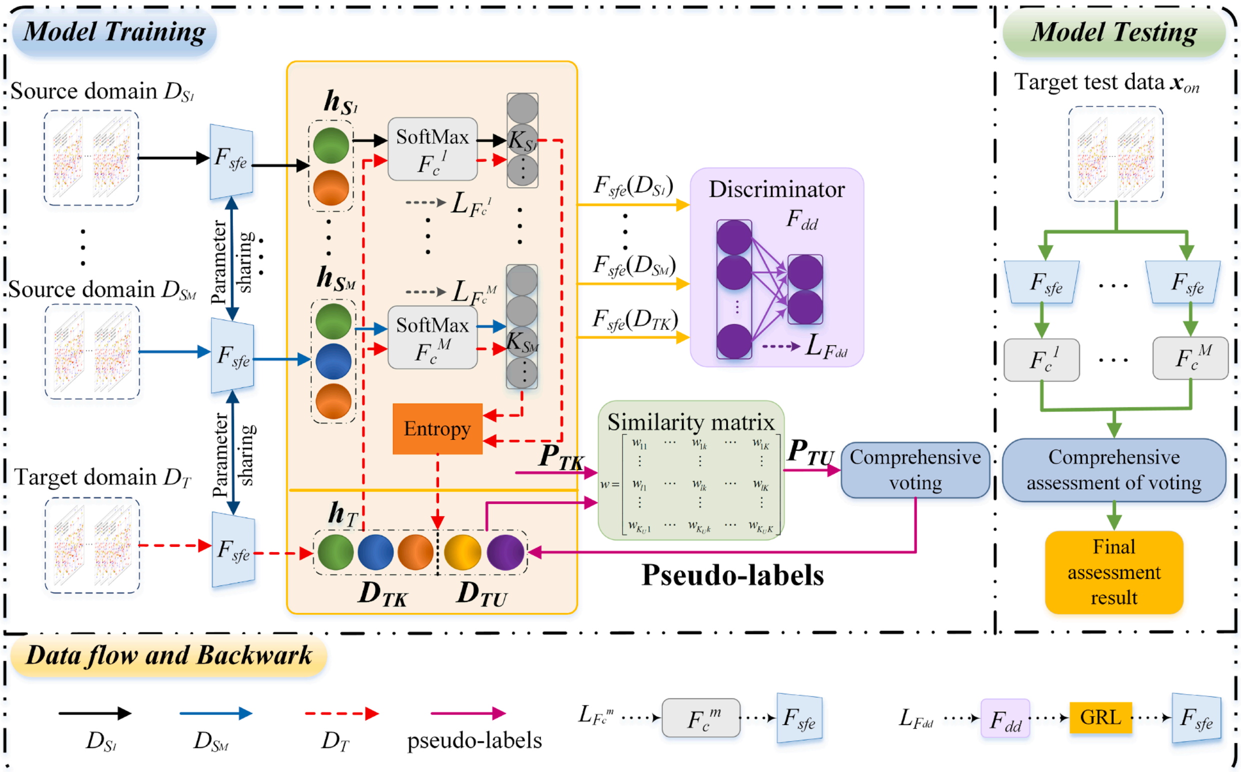


Fig. 2. Structure of the MDODTAN.

where D_k represents a dataset containing the k th performance grade of all SDs with a total of N_{D_k} samples, and $\bar{F}_{sfe,k} = (\sum_{i=1}^{N_{D_k}} F_{sfe}(\mathbf{x}_k^i))/N_{D_k}$ is the class center of the k th performance grade.

The inter-class distance is defined as follows:

$$L_{inter} = F_{mdc}(\mathbf{x}) = \sum_{q=1,q\neq p}^{K} \sum_{p=1}^{K} \left\| \bar{F}_{sfe,q} - \bar{F}_{sfe,p} \right\|^2, \tag{2}$$

where $\bar{F}_{sfe,q} = (\sum_{i=1}^{N_{D_q}} F_{sfe}(\mathbf{x}_q^i))/N_{D_q}$ and $\bar{F}_{sfe,p} = (\sum_{i=1}^{N_{D_p}} F_{sfe}(\mathbf{x}_p^i))/N_{D_p}$ are the class centers of the q th and p th performance grades, respectively.

(3) Task classifier

Based on the fact that the number of performance grades contained in each SD are different, we designed an independent task classifier for each SD, with each classifier containing a different number of nodes. In this article, the task classifier uses a FCNN as the output layer, where the activation function is SoftMax function. Taking S_m as an example, since it contains K_{S_m} performance grades, the task classifier F_c^m is set to $(K_{S_m} + 1 + K_U)$ nodes when performing the operating performance assessment task, where the first K_{S_m} nodes are used to classify all performance grades contained in S_m; the $(K_{S_m} + 1)$ th node is used to classify known performance grades that do not exist in S_m but exist in other SDs; the remaining K_U nodes are used to assess the unknown performance grades in the TD. Compared to the traditional approach of designing the same number of classifier nodes for multiple SDs, our method incorporates the actual performance grade information of each SD, significantly improving the classification accuracy of known performance grades.

(4) TD data division module

Effective separation of known and unknown performance grade data in TD is essential for establishing adversarial training between SD and TD. By inputting the TD data into the initially trained network, the preliminary assessment results of the first K known performance grades can be obtained, and the information entropy can be calculated subsequently. Taking the assessment results of TD samples in F_c^m as an example, the information entropy can be calculated as follows:

$$\zeta_{S_m}^{T,j} = H(\mathbf{a}_{S_m}^{T,j}) = -\sum_{i=1}^{K_{S_m}} a_{S_m,i}^{T,j} \log a_{S_m,i}^{T,j}, \tag{3}$$

where $\mathbf{a}_{S_m}^{T,j}$ represents the assessment result of the j th sample in the TD, and $a_{S_m,i}^{T,j}$ is the posterior probability with respect to the i th performance grade. A smaller entropy value indicates more accurate assessment results, which can then be used to partition the TD dataset.

(5) Domain discriminator

In order to reduce the data distributions difference between SD and TD, the transfer adversarial DA training is completed with F_{sfe}. In this paper, a domain discriminator with a two-layer neural network is designed. The parameters of the network nodes are 6 and 2, respectively. Through the adversarial training on known performance grades data in each source-target domain, the features of known performance grades are aligned to improve the assessment accuracy of the first K nodes of each task classifier on the known performance grades.

(6) Comprehensive assessment module

In order to avoid the situation that the known performance grades existing only in individual SDs are incorrectly evaluated as unknown performance grades due to their low weights when fusing the results of various classifiers, we propose a comprehensive voting method to fuse the assessment results of different classifiers. Only the task classifiers containing the k th known performance grade are used to calculate the posterior probability that the TD sample belongs to the k th performance grade. The synthetic posterior probability of the j th TD sample $\mathbf{x}_T^j$ with respect to the k th known performance grade is defined as follows:

$$P_k(\mathbf{x}_T^j) = \frac{1}{N_k^S} \sum_{m=1}^{N_k^S} \left[F_c^m\left(F_{sfe}(\mathbf{x}_T^j)\right) \middle| I_{k_m} \right], \tag{4}$$

where N_k^S is the number of SD containing performance grade k. I_{k_m} denotes the relative position of the k_m th performance grade in the m th task classifier. $[F_c^m(F_{sfe}(\mathbf{x}_T^j))|I_{k_m}]$ is the posterior probability of performance grade k obtained from the m th task classifier for $\mathbf{x}_T^j$.

(7) Similarity measure module

During training, F_{sfe} cannot learn the feature information of unknown performance grades in the TD, making it impossible for F_{ca} to accurately predict their labels. To solve this, we propose a pseudo-label prediction method based on the similarity matrix between unknown and known grades.

The K-Means algorithm is used to cluster unknown performance grades, and the similarities between unknown and known performance grades are measured by calculating the distance between class centers. The smaller their Euclidean distance, the greater the similarity (Yu et al., 2024). The Euclidean distance is $d_{lk} = ||\tilde{O}_l - \bar{O}_k||^2$, where $\tilde{O}_l, l = 1, 2, \ldots, K_U$ and $\bar{O}_k = \bar{F}_{sfe,k}(k = 1, 2, \ldots, K)$ represent the class centers of the unknown and known performance grades, respectively. Subsequently, the similarity between the l th unknown performance grade and the k th known performance grade is defined as $w_{lk} = \frac{1/d_{lk}}{\sum_{k=1}^{K} 1/d_{lk}}$. Therefore, the similarity matrix is denoted as $W = [w_{lk}]_{K_U \times K}$. By multiplying it with the synthetic posterior probabilities obtained through comprehensive voting method, we can further calculate the posterior probability of samples belonging to the unknown performance grade and assign pseudo-labels to them.

2.3. Offline training of the MDODTAN

The offline training process of the MDODTAN model consists four steps.

Step 1: pre-train multi-SD task classifier and shared feature extractor.

Taking S_m as an example, the training dataset D_{S_m} is input into the F_{sfe} and corresponding task classifier F_c^m, and assessment results of the first K nodes are calculated through SoftMax function as follows:

$$\hat{y}_{S_m}^{i,k} = F_c^m(y_{S_m}^i = k \Big| F_{sfe}(\mathbf{x}_{S_m}^i); \boldsymbol{\psi}_c^m) = \frac{e^{\psi_c^{m,k}} F_{sfe}(\mathbf{x}_{S_m}^i)}{\sum_{k=1}^{K} e^{\psi_c^{m,k}} F_{sfe}(\mathbf{x}_{S_m}^i)}, \tag{5}$$

where $\boldsymbol{\psi}_c^m = \{\psi_c^{m,1}, \psi_c^{m,2}, \ldots, \psi_c^{m,K}\}$ represents the parameter set of the F_c^m.

The loss function of the $F_c^m(\mathbf{h})$ is defined as follows:

$$L_{F_c^m} = -\frac{1}{n_{S_m}} \sum_{i=1}^{n_{S_m}} \sum_{k=1}^{K} y_{S_m}^{i,k} \log \hat{y}_{S_m}^{i,k}. \tag{6}$$

Therefore, to update the parameters ψ_{sfe}, ψ_{mdc} and ψ_c^m, the optimization objective can be defined as follows:

$$\min_{\psi_{sfe}, \psi_{mdc}, \psi_c^m} L_{intra} - L_{inter} + L_{F_c^m}. \tag{7}$$

Step 2: construct the known performance grade dataset of the TD and complete the multi-SD adversarial training.

According to (3), we obtain the information entropy for each target sample. Information entropy can be regarded as a measure of the effectiveness and efficiency of information (Tan et al., 2022). Generally, lower information entropy indicates more stable assessment results and higher confidence. That is to say, the TD samples with lower information entropy are more likely to belong to known performance grade dataset. Therefore, by setting the hyper-parameter threshold ζ, we construct the known performance grade dataset D_{TK} and unknown performance grade dataset D_{TU} in the TD, i.e.,

$$\begin{cases} D_{TK} \leftarrow \mathbf{x}_T^j, \text{ if } (\zeta_{S_m}^{T,j} < \zeta)_{m=1}^M \\ D_{TU} = \complement_{D_T} D_{TK} \end{cases}, \tag{8}$$

where $D_{TU} = \complement_{D_T} D_{TK}$ is the complement of D_{TK} in D_T, and the number of samples in D_{TK} and D_{TU} are denoted as $n_{D_{TK}}$ and $n_{D_{TU}}$, respectively.

After obtaining the dataset D_{TK}, a multi-SD transfer adversarial network is employed to align it with the data distributions of the known performance grades in each SD, thereby reducing the differences among different domains.

Specifically, the F_{sfe} attempts to extract domain-invariant feature representations that can deceive the F_{dd}, whereas the F_{dd} attempts to classify samples into their respective domains. Therefore, the loss function of the transfer adversarial DA network of the S_m is as follows:

$$\begin{aligned} L_{DA} &= E_{\mathbf{x},\mathbf{y} \in D_{S_m}} L_{F_c^m}\left(F_c^m\left(F_{sfe}(\mathbf{x})\right), \mathbf{y}\right) \\ &\quad - \lambda \cdot E_{\mathbf{x} \in (D_{S_m} \cup D_{TK})} L_{F_{dd}}\left(F_{dd}\left(F_{sfe}(\mathbf{x})\right), \mathbf{y}_{dd}\right), \end{aligned} \tag{9}$$

where λ is a trade-off parameter to balance the task classifier loss $L_{F_c^m}$ and the domain discrimination loss $L_{F_{dd}}$, and $L_{F_{dd}}$ is the cross-entropy loss. If $\mathbf{x} \in D_{S_m}$, its domain label is defined as $\mathbf{y}_{dd} = 1$; if $\mathbf{x} \in D_{TK}$, its domain label is defined as $\mathbf{y}_{dd} = 0$. In adversarial training, a gradient reversal layer (GRL) is used to reverse-maximize $L_{F_{dd}}$ (Ganin et al., 2016). The parameters of the MDODTAN model are updated as follows:

$$\begin{aligned} &\min_{\psi_c^m, \psi_{sfe}} L_{DA}, \\ &\min_{\psi_{dd}} L_{F_{dd}}. \end{aligned} \tag{10}$$

Step 3: make comprehensive decisions for multi-SDs and assign pseudo-labels to the TD data.

After adversarial training, we obtain the synthetic posterior probabilities of the j th sample in the TD belonging to the known performance grades are $P_{TK}^j = [P_1(\mathbf{x}_T^j), P_2(\mathbf{x}_T^j), \ldots, P_K(\mathbf{x}_T^j)]$, as shown in (4). Then, based on the similarity measure module, the similarity matrix $W = [w_{lk}]_{K_U \times K}$ is obtained. Therefore, the synthetic posterior probabilities that the TD samples belong to the unknown performance grades can be defined as follows:

$$P_{TU}^j = P_{TK}^j \cdot w^T. \tag{11}$$

At this point, we can construct the dataset $\tilde{D}_T = \{\tilde{D}_{TK}, \tilde{D}_{TU}\}$ with complete label information as follows:

$$\begin{aligned} &\tilde{D}_{TK} \leftarrow \{\tilde{\mathbf{x}}_T^j, \tilde{y}_T^j = \underset{k=1,\ldots,K}{\arg\max}(P_{TK}^j)\}_{j=1}^{n_{D_{TK}}}, \text{if } \max\{P_{TK}^j\} > \varphi, \\ &\tilde{D}_{TU} \leftarrow \{\tilde{\mathbf{x}}_T^j, \tilde{y}_T^j = \underset{k=K+1,\ldots,K+K_U}{\arg\max}(P_{TU}^j)\}_{j=1}^{n_{D_{TU}}}, \end{aligned} \tag{12}$$

where φ is the TD initial division threshold.

Step 4: retrain the shared feature extractor and the task classifier using TD data.

After completing the training, the posterior probabilities of each task classifier are fused by comprehensive voting method, and then more accurate synthetic posterior probabilities $\hat{P}_T^j$ of the TD data can be obtained. The posterior probability $P_k(\tilde{\mathbf{x}}_T^j)$ of known performance grade can be obtained in (4). The posterior probability of unknown performance grade can be calculated as follows:

$$P_{K+l}(\tilde{\mathbf{x}}_T^j) = \frac{1}{M} \sum_{m=1}^{M} \left[F_c^m\left(F_{sfe}(\tilde{\mathbf{x}}_T^j)\right) | I_{(K+l)_m}\right], \tag{13}$$

where $l \in K_U$, $[F_c^m(F_{sfe}(\tilde{\mathbf{x}}_T^j)) | I_{(K+l)_m}]$ represents the posterior probability of performance grade $(K + l)$ obtained by F_c^m, i.e., the posterior probability with respect to the l th unknown performance grade in the TD. Therefore, we can get $\hat{P}_T^j = [P_1(\tilde{\mathbf{x}}_T^j), \ldots, P_K(\tilde{\mathbf{x}}_T^j), \ldots, P_{K+K_U}(\tilde{\mathbf{x}}_T^j)]$.

According to the comprehensive assessment results, the samples of TD and their corresponding pseudo-labels are further updated. The updated TD dataset is recorded as follows:

$$\tilde{\tilde{D}}_T \leftarrow \{\tilde{\mathbf{x}}_T^j, \tilde{y} = \underset{k=1,\ldots,K+K_U}{\arg\max}(\hat{P}_T^j)\}_{j=1}^{n_T}. \tag{14}$$

Then $\tilde{\tilde{D}}_T$ is divided into $(K + K_U)$ datasets $\tilde{\tilde{D}}_{T,1}, \ldots, \tilde{\tilde{D}}_{T,K}, \ldots, \tilde{\tilde{D}}_{T,K+K_U}$ according to performance grade labels, and the sample numbers of each dataset are guaranteed to be balanced. The known performance grade datasets $\tilde{\tilde{D}}_{T,1}, \ldots, \tilde{\tilde{D}}_{T,K}$ are added to the SD that contains these performance grades, and the unknown performance grade datasets $\tilde{\tilde{D}}_{T,K+1}, \ldots, \tilde{\tilde{D}}_{T,K+K_U}$ are added to each SD. Then the training process of step 1 is repeated, and the final assessment result is obtained through comprehensive voting. The offline training process is shown in Algorithm 1.

In summary, the overall learning objective for MDODTAN is

$$L_{total} = L_{intra} - \gamma_1 L_{inter} + \gamma_2 \sum_{m=1}^{M} L_{F_c^m} - \gamma_3 L_{F_{dd}}, \tag{15}$$

where γ_1, γ_2 and γ_3 are trade-off parameters. The training process for this objective can be decomposed into the following two parts.

$$(\hat{\psi}_{sfe}, \hat{\psi}_{mdc}, \hat{\psi}_c^m) = \arg\min_{\psi} L_{total},$$
$$(\hat{\psi}_{dd}) = \arg\min_{\psi} L_{F_{dd}}. \tag{16}$$

It is important to note that ψ_{sfe} is optimized to minimize L_{intra}, maximize L_{inter}, minimize $\sum_{m=1}^{M} L_{F_c^m}$, and maximize $L_{F_{dd}}$ via adversarial training. This adversarial training between ψ_{sfe} and ψ_{dd} encourages F_{sfe} to learn domain-invariant feature representations. The key innovation in MDODTAN lies in the gradient fusion mechanism that combines gradients from multiple structurally different task classifiers. Each SD $S_m, m = 1, 2, \ldots, M$ has its own task classifier F_c^m with different output dimensions, reflecting the varying number of performance grades in each domain. During training, these classifiers generate gradients that are fused through the weighted sum term $\gamma_2 \sum_{m=1}^{M} L_{F_c^m}$ in Eq. (15).

For the optimization problem in Eq. (16), the stochastic gradient descent (SGD) algorithm updates the network parameters as follows:

$$\psi_{sfe} \leftarrow \psi_{sfe} - \alpha \cdot \left[\frac{\partial L_{total}}{\partial \psi_{sfe}}\right],$$
$$\psi_{mdc} \leftarrow \psi_{mdc} - \alpha \cdot \left[\frac{\partial L_{\text{intra}}}{\partial \psi_{mdc}} - \gamma_1 \frac{\partial L_{\text{inter}}}{\partial \psi_{mdc}}\right],$$
$$\psi_c^m \leftarrow \psi_c^m - \alpha \cdot \left[\gamma_2 \frac{\partial L_{F_c^m}}{\partial \psi_c^m}\right], \quad m = 1, \ldots, M,$$
$$\psi_{dd} \leftarrow \psi_{dd} - \alpha \cdot \left[\gamma_3 \frac{\partial L_{F_{dd}}}{\partial \psi_{dd}}\right], \tag{17}$$

where α is the learning rate.

Algorithm 1 The offline training of the MDODTAN.

Input: SD datasets $D_{S_m}, m = 1, \ldots, M$, and TD datasets D_T. Initialize network parameters.
1: **for** number of training epoch **do**
2: Compute the intra-class distance L_{intra}, inter-class distance L_{inter} and the task classifier loss function $L_{F_c^m}$.
3: Divide the TD dataset into D_{TK} and D_{TU}.
4: Compute the transfer adversarial DA loss function L_{DA}.
5: Compute the synthetic posterior probabilities P_{TK}^j and P_{TU}^j of unlabeled TD data.
6: Update the TD dataset with complete label information: $\tilde{D}_T = \{\tilde{D}_{TK}, \tilde{D}_{TU}\}$.
7: Using labeled $\tilde{D}_T$ to obtain more accurate TD dataset $\tilde{\tilde{D}}_T$ with labeled information.
8: Divide $\tilde{\tilde{D}}_T$ into $(K + K_U)$ datasets and add them to each SD, repeat step 1.
9: Update parameters ψ_{sfe}, ψ_{mdc}, ψ_c^m and ψ_{dd} by (17).
10: **end for**
Output: MDODTAN model.

2.4. Online assessment based on MDODTAN

For the online data $\mathbf{x}_{on}$, combined with the task classifier, the following comprehensive assessment results based on MDODTAN are obtained:

$$\hat{y}_{on}^k = F_c(\mathbf{y}_{on} = k \middle| F_{sfe}(\mathbf{x}_{on}); \tilde{\psi}_c) = \frac{e^{\tilde{\psi}_c^k} F_{sfe}(\mathbf{x}_{on})}{\sum_{k=1}^{K+K_U} e^{\tilde{\psi}_c^k} F_{sfe}(\mathbf{x}_{on})}, \tag{18}$$

where $\hat{\mathbf{y}}_{on} = \left[\hat{y}_{on}^1, \ldots, \hat{y}_{on}^K, \ldots, \hat{y}_{on}^{K+K_U}\right]^T$ is the output from the task classifier. The synthetic posterior probabilities $\hat{P}_{on} = [P_1(\mathbf{x}_{on}), \ldots, P_K(\mathbf{x}_{on}), \ldots, P_{K+K_U}(\mathbf{x}_{on})]$ are obtained by synthesizing the posterior probabilities as in (13). The index corresponding to the maximum synthetic posterior probability is the current operating performance grade, i.e.,

$$\hat{k}_{on} = \arg \max_{k=1,\ldots,K+K_U} \{\hat{P}_{on}\}. \tag{19}$$

Table 2
Average current of different EFMF.

Current	1-Optimal	2-Suboptimal	3-General	4-Poor	5-Worse
S_1	15000 ± 10	14800 ± 15	15200 ± 15	15025 ± 35	15030 ± 45
S_2	14900 ± 10	14700 ± 15	15100 ± 15	14925 ± 35	14930 ± 45
TD	15200 ± 10	14970 ± 15	15300 ± 15	15210 ± 35	15215 ± 45

Table 3
Performance grades included in each domain.

Domain	S_1	S_2	TD
Performance Grades	{1,3}	{1,2,3}	{1,2,3,4,5}

3. Experimental verification

3.1. Process description and data collect

In the melting process, EFMF uses three-phase electrodes to generate large arc currents and melt magnesite ore into liquid MgO. As the heat is absorbed, impurities with different melting points and densities gradually precipitate and form high-purity MgO crystals. To maintain product quality, current can be adjusted based on raw material particle size or melting point. Higher product quality and economic benefits indicate better performance grade. Otherwise, performance grade may deviate from optimal performance grade. Based on product quality, economic indicators, and expert experience, performance grade is divided into five grades: optimal (1), suboptimal (2), general (3), poor (4), and worse (5). In this study, grades 1, 2, and 3 represent known performance grades for both SD and TD, while grades 4 and 5 represent unknown performance grades of TD.

To collect current data, a mechanism model of the magnesia smelting process is established, and current data under different performance grades in different smelting processes can be obtained (Niu et al., 2022). Based on expert experience, the average current at the optimal performance grade is 15000A. The average current data of each SD and TD are shown in Table 2. According to the setting of open-set operating performance assessment, the performance grades of each SD and TD are designed as shown in Table 3.

During offline training, each performance grade should have an equal number of samples. In this study, each performance grade contains 800 training samples. The test set contains 1000 TD samples, with 200 samples per performance grade. All SD data are labeled, while TD data are unlabeled. The data distributions of each SD and TD are different.

3.2. Determination of hyper-parameters

Through cross validation, the relevant parameters are given as follows: learning rate $\alpha = 0.001$, trade-off parameters $\gamma_1 = \gamma_2 = 0.1$, $\gamma_3 = 0.5$, information entropy threshold $\zeta = 0.1$ and TD initial division threshold $\varphi = 0.97$. There are two important hyper-parameters ζ and φ. ζ is used to effectively separate the known and unknown performance grade datasets in the TD, and φ is used to initially assign pseudo-labels to the TD data. In our experiment, the values of two parameters are limited to $\zeta \in [0, 1.6]$ and $\varphi \in (0, 1)$, respectively. Their influence on model performance is shown in Fig. 3. From Fig. 3(a), when $\zeta \in [0.1, 0.6]$, it can effectively divide the TD dataset, and the model reaches high accuracy. Further increasing ζ reduces partition accuracy. One possible reason is

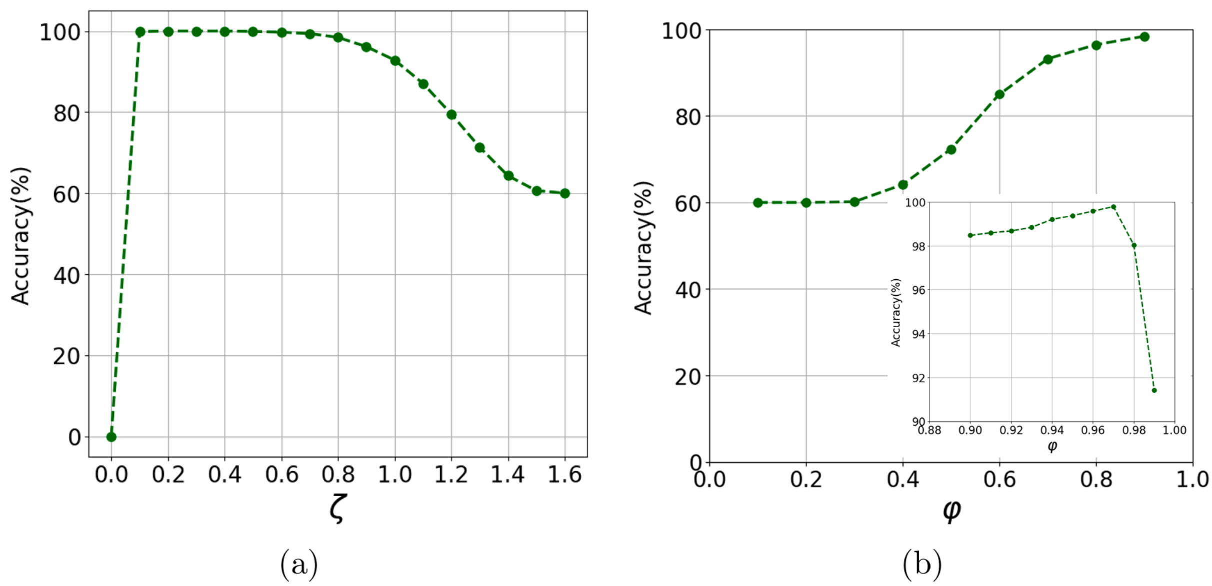


Fig. 3. The impact of hyper-parameters. (a) ζ. (b) φ.

that higher entropy means that the assessment results are uncertain and less reliable, so we set $\zeta = 0.1$. On the other hand, Fig. 3(b) shows that when $\varphi = 0.97$, the accuracy of known performance grades in the TD being correctly assigned pseudo-labels reaches the maximum.

3.3. Experimental results

1) Feature visualization: We use t-distributed stochastic neighbor embedding (t-SNE) (van der Maaten & Hinton, 2008) to visualize the features after adversarial training of S_1 and S_2. Fig. 4 shows the feature distributions for each known performance grade in the SD and TD. By adversarial training, features belonging to different performance grades are separated, and the deep feature distributions of each performance grade across SD and TD are almost the same. The results show that by using the MDODTAN model, we can effectively enrich and extend our training set by using the labeled data in each historical magnesia smelting, so as to further improve the DA task and the assessment accuracy of the target task. Furthermore, the task classifier loss and domain discriminator loss after adversarial training between SD and TD are shown in Fig. 5. It can be seen that the loss value approaches 0 with about 400 iterations.

Table 4
Confusion matrices for performance comparison.

(a) MDODTAN method

Grades		Predicted grades				
		1	2	3	4	5
Actual grades	1	0.99	0	0	0	0.01
	2	0	1	0	0	0
	3	0	0	0.995	0.005	0
	4	0	0	0	0.985	0.015
	5	0	0	0	0	1

(b) Cluster-based method

Grades		Predicted grades				
		1	2	3	4	5
Actual grades	1	0.995	0.005	0	0	0
	2	0	1	0	0	0
	3	0	0	1	0	0
	4	0	0	0	0.515	0.485
	5	0	0	0	0.275	0.725

2) Analysis of computational cost: The proposed MDODTAN method in this paper significantly improves the accuracy of cross-domain performance assessment by designing an independent classifier for each SD and conducting pairwise adversarial training. However, this inevitably increases the computational overhead during the training phase. The main sources of overhead stem from three aspects: the parameters of the independent classifiers increase linearly with the number of SDs M, adversarial training is required M times between each SD and the TD, and clustering computations and iterative optimization of pseudo-labels are necessary for K_U unknown performance grades. To highlight the effectiveness of the multi-SD independent transfer adversarial strategy, we conducted a comparative experiment by merging S_1 and S_2 into a larger SD S. While this approach reduces the complexity of the adversarial training phase from $O(M)$ to $O(1)$, the feature visualization results in Fig. 6(a) show that after adversarial training, although each known performance grade in S and TD has been aligned, the performance grade alignment is poor, with a smaller distance between the grades "optimal" and "suboptimal." Additionally, in Fig. 6(b), the loss value only approaches 0 after about 600 iterations, and the convergence speed is significantly slower than in Fig. 5. This demonstrates that, although the independent adversarial mechanism incurs higher computational costs, it is crucial for preserving feature discriminability, facilitating knowledge transfer, and achieving high-precision classification.

3) Effect of similarity matrix: To demonstrate the necessity of constructing a similarity matrix for assigning pseudo-labels to unknown samples, we directly applied traditional K-means clustering on the target data. The confusion matrix is shown in Table 4. The results indicate that the classification accuracy of unknown samples is 62%, which is significantly lower than the 99.25% achieved by our method. This is because clustering analysis is sensitive to noise and outliers, which can lead to misclassification, especially at class boundaries or in dense regions. In contrast, our proposed method of constructing the similarity matrix incorporates knowledge from SD, expert experience, and unsupervised clustering, effectively improving the accuracy and reliability of the classification.

4) Sensitivity analysis of K_U: To investigate the performance of MDODTAD in different settings of K_U, we adopted three evaluation metrics: the accuracy for known classes classification (Acc_{kwn}), the accuracy for unknown classes classification (Acc_{unk}), and the overall accuracy (Acc). This study focuses on evaluating the model's sensitiv-

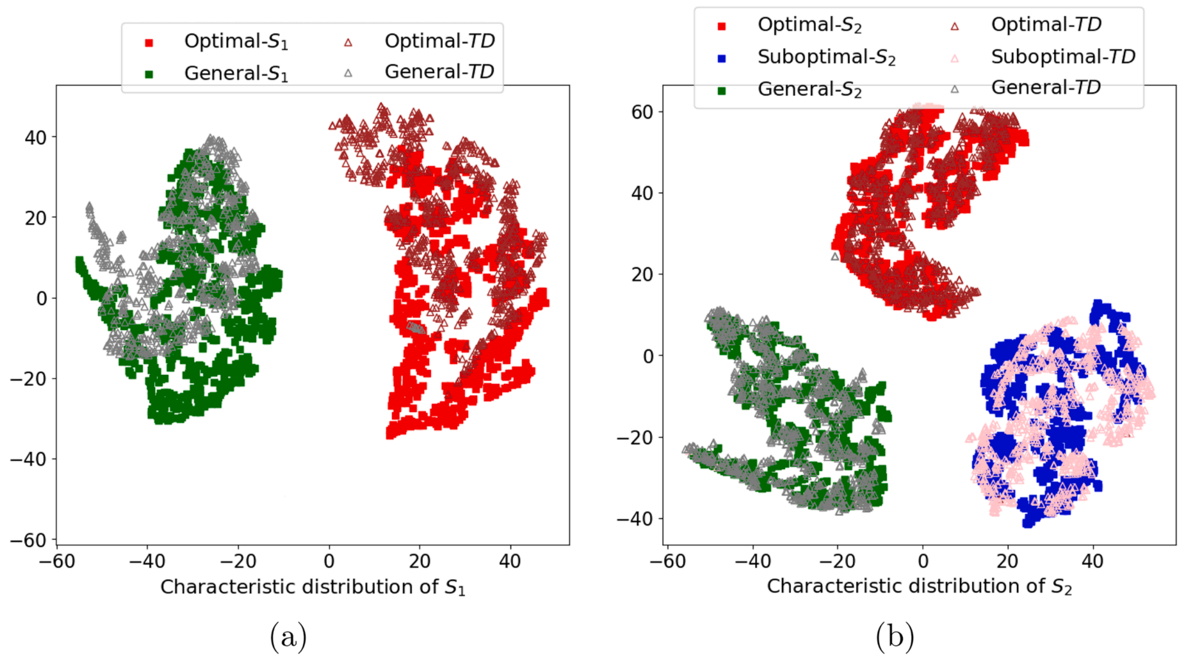


Fig. 4. The T-SNE feature visualization results.

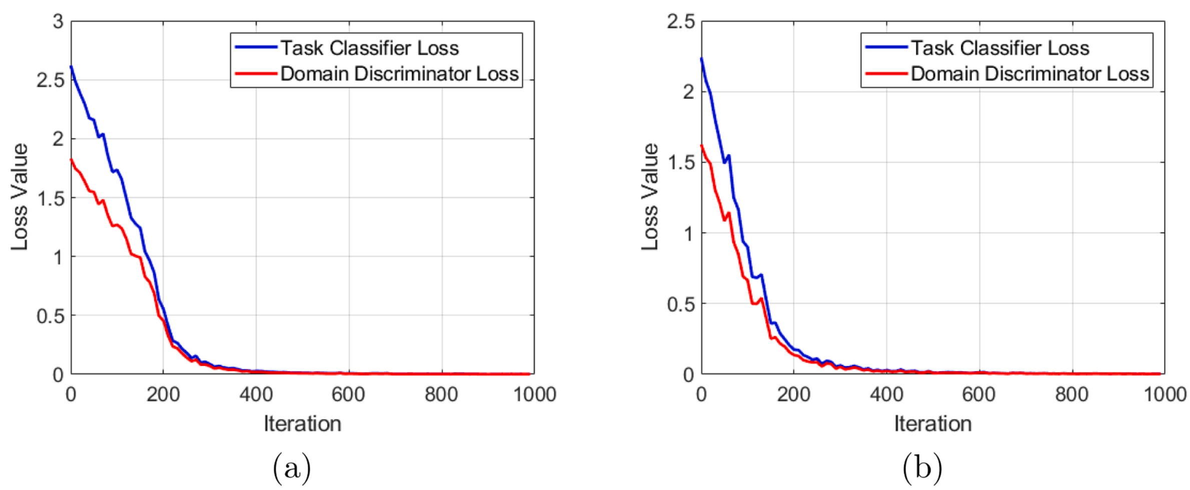


Fig. 5. The losses after adversarial training. (a) S_1 and D_{TK}. (b) S_2 and D_{TK}.

ity when K_U varies from 1 to 4, as shown in Fig. 7. The results indicate that when K_U is set between 1 and 3, the model consistently achieves high performance across all metrics, with overall accuracy exceeding 95%. However, when $K_U = 4$, all performance metrics decline significantly, particularly the accuracy for unknown classes classification, which drops to 42.25%. This demonstrates that the MDODTAN method maintains strong robustness when K_U is slightly overestimated or underestimated, largely due to the fault-tolerant capability of the comprehensive voting method and similarity matrix, which enables adaptive adjustment to structural deviations caused by misestimation of K_U. Notably, when $K_U = 4$, the similarity matrix fails to assign meaningful labels to unknown classes, forcing the model to rely solely on clustering methods, leading to a sharp drop in performance. These experimental results strongly demonstrate that, as long as expert knowledge can provide a reasonable range for K_U, the method is not critically sensitive to the exact value of this parameter. This finding establishes a solid foundation for deploying the method in practical industrial applications.

3.4. Comparisons with other methods

Several different DA methods are used for comparison: DANN (Ganin et al., 2016), OSDAB (Saito et al., 2018), STA (Liu et al., 2019), MADTN (Zuqiang et al., 2023), UDA-ROT (Zeng et al., 2025). For a fair comparison, the feature extractors, classifiers and domain discriminator of the different comparison methods have the same network structure as our method. DANN is a classical close-set DA method, which cannot classify unknown classes. OSDAB, STA and MADTN are OSDA methods, which can group all unknown classes into a single category. UDA-ROT can distinguish unknown classes into different private classes. The online assessment results based on different methods are shown in Fig. 8. It can be seen from Fig. 8 that the proposed MDODTAN method has higher

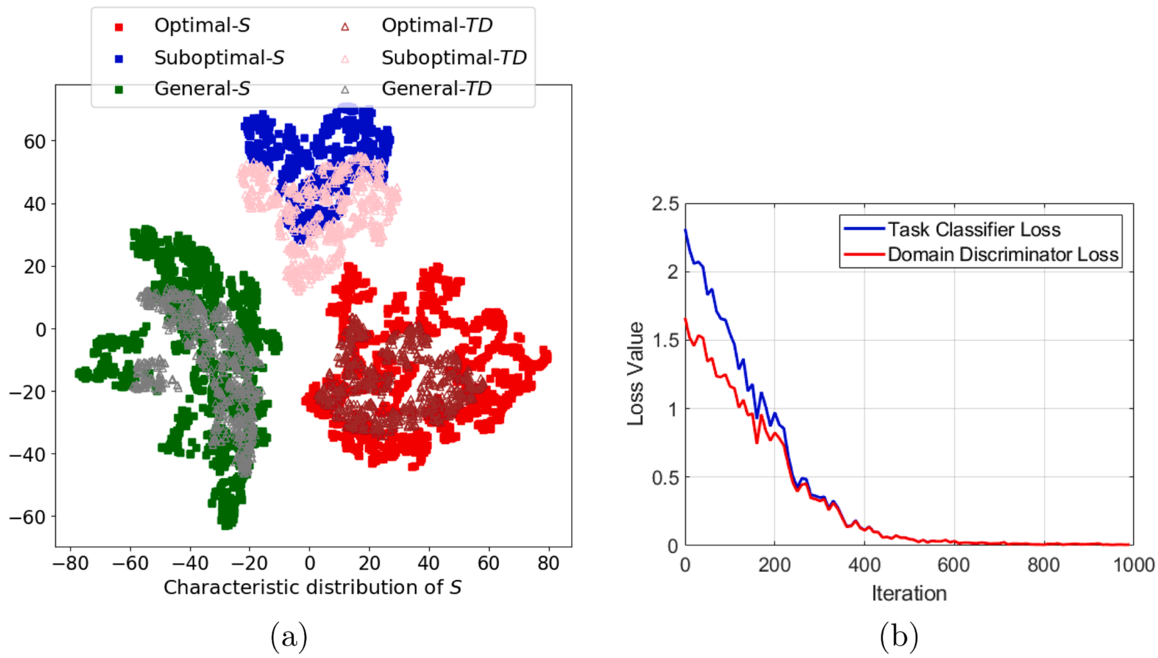


Fig. 6. Visualization result and loss value.

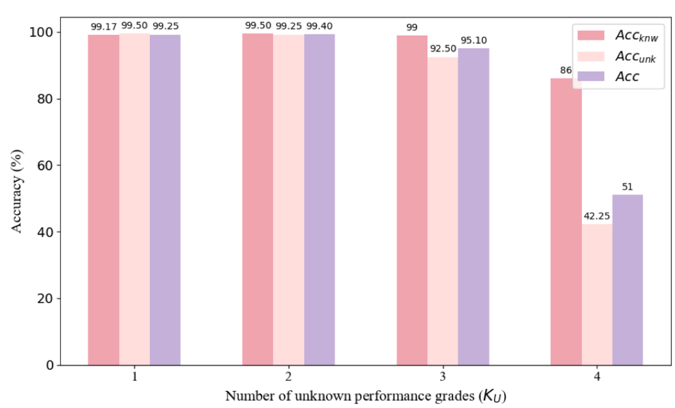


Fig. 7. Analysis of the performance in different K_U settings.

Table 5
Evaluation metrics for different methods.

Method	Grade	Precision(%)	Recall(%)	F1-scores(%)
DANN (Ganin et al., 2016)	1	70.42	**100**	82.46
accuracy = 60%	2	49.88	**100**	66.56
	3	63.49	**100**	77.67
	unknown	0	0	0
OSDAB (Saito et al., 2018)	1	95.22	99.50	97.31
accuracy = 86.2%	2	80.00	**100**	88.89
	3	71.94	**100**	83.68
	unknown	**100**	65.75	79.34
STA (Liu et al., 2019)	1	96.62	**100**	98.28
accuracy = 88.2%	2	71.94	**100**	83.68
	3	85.84	**100**	92.38
	unknown	**100**	70.50	82.70
MADTN (Zuqiang et al., 2023)	1	97.56	**100**	98.77
accuracy = 91.2%	2	72.99	**100**	84.39
	3	95.69	**100**	97.80
	unknown	**100**	78.00	87.64
UDA-ROT (Zeng et al., 2025)	1	**100**	99.50	**99.75**
accuracy = 97.8%	2	**100**	**100**	**100**
	3	**100**	**100**	**100**
	4	**100**	89.50	94.46
	5	90.09	**100**	94.79
MDODTAN	1	**100**	99.00	99.50
accuracy = **99.4%**	2	**100**	**100**	**100**
	3	**100**	99.50	99.75
	4	99.49	**98.50**	**98.99**
	5	**97.56**	**100**	**98.77**

accuracy in assessment results, fewer false positives, and separates all unknown performance grades.

In order to further compare the performance of various methods from a quantitative perspective, Table 5 provides the evaluation metrics for different methods. It is found that all methods have almost the same performance on the assessment accuracy of the known performance grade. Additionally, the accuracy of the comparison methods ranges from 60% to 97.8%, which is significantly lower than the 99.4% achieved by our method. In terms of overall performance, while both MDODTAN and UDA-ROT are capable of fine-grained partitioning of unknown classes, MDODTAN achieves a superior overall accuracy of 99.4%, representing a 1.6 percentage point improvement over UDA-ROT (97.8%). A detailed examination of the Precision, Recall, and F1-scores for individual performance grades in Table 5 reveals that MDODTAN's advantage is particularly pronounced in the classification of unknown classes (grades 4 and 5). For instance, in grade 4, MDODTAN attains a Recall of 98.5%, significantly outperforming UDA-ROT's 89.5%. For grade 5, MDODTAN achieves a perfect Precision of 97.56% and an F1-score of 98.77%, again surpassing its counterpart. This demonstrates that MDODTAN significantly enhances the ability to recognize unknown classes while maintaining high classification performance on known classes, a point that is particularly important in open-set assessment. In practical applications of industrial POPA, especially in high-energy consumption production processes like EFMF, even a 1% performance improvement can lead to significant economic benefits. The multi-SD independent adversarial training and the fine-grained unknown class segmentation mechanism based on comprehensive voting method and similarity matrices proposed by MDODTAN provide a more interpretable, stable, and practical solution for open-set POPA. Furthermore, to demonstrate the significant

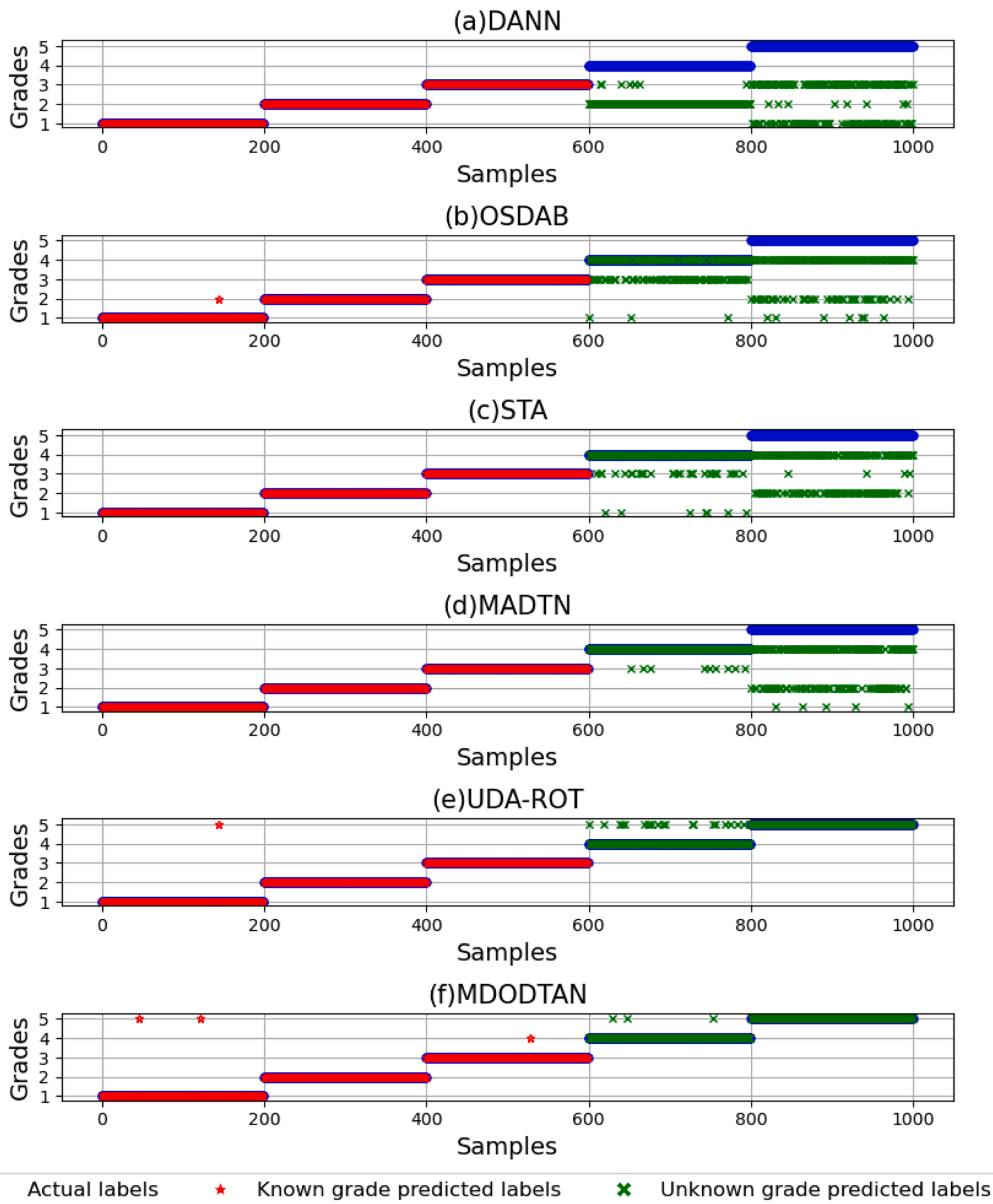


Fig. 8. The online assessment results of different methods.

differences between the two methods, a Welch's independent samples t-test (Welch, 1947) is conducted on the 10-fold cross-validation results. The test revealed MDODTAN achieved a significantly higher accuracy (99.4% ± 0.15%) compared to UDA-ROT (97.8% ± 0.45%) with the test statistic $t(10.93) = 10.63$ and a highly significant $p = 1.8 \times 10^{-7} \ll 0.01$). This result provides strong evidence that the observed improvement is substantive and not due to random chance.

To demonstrate the excellent robustness and stability of the MDODTAN method in multi-classifier fusion, all methods undergo 10-fold cross-validation. Fig. 9 presents a box plot comparing the classification accuracy of six DA methods for the POPA task of the EFMF smelting process. Outliers, denoted by hollow circles, indicate data points that lie far from the overall data distribution. As shown in the figure, the proposed MDODTAN method demonstrates outstanding performance. It achieves an accuracy of 99.4%, with a highly concentrated distribution and no outliers, indicating that the method not only attains the highest classification accuracy but also exhibits remarkable stability. In contrast, while the UDA-ROT method achieves sub-optimal performance, its box plot range is wider compared to MDODTAN, reflecting relatively insufficient stability. The performance of the MADTN, STA, and OSDAB methods decreases sequentially. The traditional DANN method, serving as a baseline, consistently maintains an accuracy of 60% due to its inability to handle unknown classes. Leveraging its multi-source domain independent adversarial training mechanism and comprehensive voting strategy, MDODTAN achieves nearly perfect performance assessment in open-set scenarios, demonstrating strong robustness and stability.

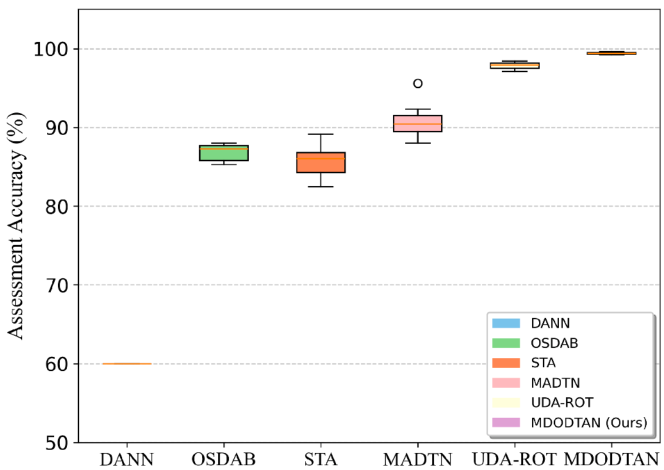


Fig. 9. Box plot for different methods.

4. Conclusions

This paper proposes a novel MDODTAN model, aiming to address the problem of multi-source open-set POPA. By minimizing intra-class distance and maximizing inter-class distance, the MDODTAN method effectively reduces the differences between data from each performance grade, thereby enhancing discriminability. Simultaneously, different task classifiers are designed according to the characteristics of each SD, enabling accurate classification of all known performance grades. Furthermore, domain-invariant features are extracted through a multi-SD adversarial network, which further clarifies the distinction between known and unknown performance grades. Combined with a comprehensive voting mechanism and a TD division threshold, the performance assessment of the model is improved, and unknown performance grades can be accurately classified and evaluated. Finally, the effectiveness and potential of the MDODTAN method in addressing this type of problem are validated through simulated experiments on the melting process of EFMF.

It is important to clarify that this study uses simulated data generated based on an EFMF mechanistic model. Although the simulation results validate the effectiveness of the method, its practical industrial application may face multiple challenges such as real-world data noise, incompleteness, and process time-variability. To address these challenges, future work will focus on exploring robust data preprocessing and adaptive updating strategies, as well as seeking to validate and optimize the method using actual production data. Additionally, technologies such as dynamic source domain selection, lightweight design, and incremental updates will be employed to enhance the scalability of the method in large-scale multi-source scenarios, thereby reducing computational costs and promoting its practical application in complex industrial settings.

CRediT authorship contribution statement

Yan Liu: Writing – review & editing, Methodology, Funding acquisition, Conceptualization; **Lulu Fu:** Writing – original draft, Validation, Software, Methodology, Conceptualization; **Yulu Xiong:** Validation, Software; **Siqi Wang:** Validation, Software, Conceptualization; **Fei Chu:** Supervision, Conceptualization; **Chenhui Bao:** Validation, Supervision; **Fuli Wang:** Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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